Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Roestar Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
10407 W Bradbury RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X026-X003-XXXX

B. OPERATORS

Roest, Arie

Operator name: Roest, Arie

Telephone no.: (209) 632-6464 Landline, (209) 648-3930 Cellular

Mailing Address Number and Street	City	State	Zip Code
3812 S Morgan RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Roest, Arie

Legal owner name: Roest, Arie

Telephone no.: (209) 632-6464 Landline, (209) 648-3930 Cellular

Mailing Address Number and Street	City	State	Zip Code
3812 S Morgan RD	Turlock	CA	95380

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	305	279	0	0
Number under roof	1,019	0	0	0	0	0
Maximum number	1,019	0	307	281	0	0
Average number	1,019	0	307	281	0	0
Avg live weight (lbs)	1,400	0	900	770		

Predominant milk cow breed: Holstein

Average milk production: 76 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 32,694.33 *tons per reporting period*

Total nitrogen from manure: 406,935.21 *lbs per reporting period*

After ammonia losses (30% loss applied): 284,854.65 *lbs per reporting period*

Total phosphorus from manure: 69,024.93 *lbs per reporting period*

Total potassium from manure: 199,164.80 *lbs per reporting period*

Total salt from manure: 479,796.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 37,710,283 *gallons*

Total nitrogen generated: 137,087.24 *lbs*

Total phosphorus generated: 36,740.85 *lbs*

Total potassium generated: 213,928.94 *lbs*

Total salt generated: 1,222,431.78 *lbs*

	37,710,283	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	37,710,283	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well #1	Ground water
Domestic Well - Well #2	Ground water
Domestic Well - Well #3	Ground water
Domestic Well - Well #4	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	19	19	2	process wastewater	X058-X027-X009-XXXX
2	20	20	2	process wastewater	X058-X026-X002-XXXX
3	49	49	2	process wastewater	X058-X026-X003-XXXX
4	92	92	2	process wastewater	X058-X026-X003-XXXX X058-X026-X007-XXXX X058-X026-X008-XXXX
5	76	76	2	process wastewater	X058-X025-X002-XXXX
6	34	34	2	process wastewater	X058-X025-X001-XXXX
Totals for areas that were used for application	290	290	12		
Totals for areas that were not used for application					
Land application area totals	290	290	12		

B. CROPS AND HARVESTS

1

Field name: 1

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	211.00 *ton*	Dry-weight		57.2	22,800.00	3,700.00	28,200.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.11	216.74	35.17	268.07	1,064.68

1

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	579.00 *ton*	Dry-weight		70.8	12,700.00	2,700.00	24,900.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.47	226.02	48.05	443.14	1,370.34

2

Field name: 2

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	247.00 *ton*	Dry-weight		57.5	20,700.00	3,500.00	29,300.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.35	217.30	36.74	307.58	1,165.22

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	619.00 *ton*	Dry-weight		70.5	13,700.00	2,500.00	25,400.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.95	250.17	45.65	463.82	1,424.32

3

Field name: 3

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	542.00 *ton*	Dry-weight		60.6	22,900.00	3,600.00	30,500.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.06	199.60	31.38	265.85	984.94

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	1,439.00 *ton*	Dry-weight		70.0	13,800.00	2,600.00	24,400.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.37	243.16	45.81	429.94	1,303.91

4

Field name: 4

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 92 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	1,041.00 *ton*	Dry-weight		64.0	23,400.00	3,900.00	31,600.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.32	190.64	31.77	257.44	896.17

4

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	2,739.00 *ton*	Dry-weight		69.3	12,600.00	2,600.00	21,900.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.77	230.33	47.53	400.33	1,169.91

5

Field name: 5

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 76 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	636.00 *ton*	Dry-weight		62.9	22,100.00	3,800.00	31,900.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.37	137.23	23.60	198.08	887.94

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	2,311.00 *ton*	Dry-weight		70.4	10,300.00	2,800.00	25,300.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.41	185.42	50.40	455.44	1,404.11

6

Field name: 6

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	398.00 *ton*	Dry-weight		56.8	19,700.00	3,500.00	28,200.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.71	199.24	35.40	285.21	1,051.84

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	1,023.00 *ton*	Dry-weight		70.3	13,000.00	2,700.00	24,400.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.09	232.34	48.26	436.09	1,340.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/07/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.50	26.96	192.00	1,028.49	495,071.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.44	3,569,550.00 *gal*
Application event totals		103.50	26.96	192.00	1,084.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.78	26.77	190.66	1,021.32	491,617.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.63	3,771,600.00 *gal*
Application event totals		102.78	26.77	190.66	1,080.95	

1 - 05/18/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.25	3,367,500.00 *gal*
Application event totals		0.00	0.00	0.00	53.25	

1 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.79	2,896,050.00 *gal*
Application event totals		0.00	0.00	0.00	45.79	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.89	20.22	106.18	629.37	400,266.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.26	2,862,375.00 *gal*
Application event totals		73.89	20.22	106.18	674.63	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.76	20.73	108.88	645.34	410,426.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.79	2,896,050.00 *gal*
Application event totals		75.76	20.73	108.88	691.13	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.94	21.05	110.57	655.35	416,789.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.94	2,842,170.00 *gal*
Application event totals		76.94	21.05	110.57	700.29	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.41	20.91	109.80	650.81	413,900.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.79	2,896,050.00 *gal*
Application event totals		76.41	20.91	109.80	696.60	

1 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.79	2,896,050.00 *gal*
Application event totals		0.00	0.00	0.00	45.79	

2 - 11/07/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.95	26.04	185.42	993.25	503,268.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.58	3,367,500.00 *gal*
Application event totals		99.95	26.04	185.42	1,043.83	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.11	25.82	183.85	984.82	498,998.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.57	3,300,150.00 *gal*
Application event totals		99.11	25.82	183.85	1,034.39	

2 - 05/18/2023: Corn, silage

Field name: 2

Crop: Corn, silage Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.09	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	51.09	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.52	3,030,750.00 *gal*
Application event totals		0.00	0.00	0.00	45.52	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.65	23.16	121.64	720.98	482,665.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.02	2,997,075.00 *gal*
Application event totals		84.65	23.16	121.64	766.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.03	21.35	112.14	664.68	444,972.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.50	2,896,050.00 *gal*
Application event totals		78.03	21.35	112.14	708.18	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.63	11.41	138.21	648.98	448,233.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.02	2,997,075.00 *gal*
Application event totals		86.63	11.41	138.21	694.00	

2 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.57	21.22	111.47	660.68	442,293.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.50	2,896,050.00 *gal*
Application event totals		77.57	21.22	111.47	704.18	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.01	2,929,725.00 *gal*
Application event totals		0.00	0.00	0.00	44.01	

3 - 11/07/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.23	23.77	169.24	906.56	1,125,401.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.90	7,812,600.00 *gal*
Application event totals		91.23	23.77	169.24	954.46	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.31	24.57	174.95	937.18	1,163,403.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.69	7,778,925.00 *gal*
Application event totals		94.31	24.57	174.95	984.87	

3 - 05/18/2023: Corn, silage

3 - 05/18/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.76	8,115,675.00 *gal*
Application event totals		0.00	0.00	0.00	49.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.45	7,576,875.00 *gal*
Application event totals		0.00	0.00	0.00	46.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.34	21.71	114.01	675.77	1,108,378.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.66	7,610,550.00 *gal*
Application event totals		79.34	21.71	114.01	722.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.21	21.67	113.83	674.67	1,106,561.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.22	7,374,825.00 *gal*
Application event totals		79.21	21.67	113.83	719.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.89	21.59	113.37	671.96	1,102,124.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.01	7,341,150.00 *gal*
Application event totals		78.89	21.59	113.37	716.97	

3 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.33	21.70	114.00	675.67	1,108,211.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.22	7,374,825.00 *gal*
Application event totals		79.33	21.70	114.00	720.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.80	7,307,475.00 *gal*
Application event totals		0.00	0.00	0.00	44.80	

4 - 11/07/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.86	23.41	166.69	892.93	2,081,219.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.70	15,221,100.00 *gal*
Application event totals		89.86	23.41	166.69	942.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.75	23.38	166.50	891.88	2,078,763.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.69	15,524,175.00 *gal*
Application event totals		89.75	23.38	166.50	942.57	

4 - 05/18/2023: Corn, silage

4 - 05/18/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.72	14,614,950.00 *gal*
Application event totals		0.00	0.00	0.00	47.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.83	14,648,625.00 *gal*
Application event totals		0.00	0.00	0.00	47.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.29	21.70	113.95	675.38	2,079,817.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.05	14,715,975.00 *gal*
Application event totals		79.29	21.70	113.95	723.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.54	20.39	107.11	634.87	1,955,088.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.61	14,581,275.00 *gal*
Application event totals		74.54	20.39	107.11	682.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.35	21.44	112.59	667.35	2,055,086.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.83	14,648,625.00 *gal*
Application event totals		78.35	21.44	112.59	715.18	

4 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.88	20.76	109.05	646.36	1,990,444.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.27	14,783,325.00 *gal*
Application event totals		75.88	20.76	109.05	694.63	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.05	14,715,975.00 *gal*
Application event totals		0.00	0.00	0.00	48.05	

5 - 11/07/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.01	22.17	116.42	690.02	1,755,359.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.18	12,695,475.00 *gal*
Application event totals		81.01	22.17	116.42	740.20	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.14	22.20	116.60	691.11	1,758,133.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.45	12,257,700.00 *gal*
Application event totals		81.14	22.20	116.60	739.56	

5 - 05/18/2023: Corn, silage

5 - 05/18/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.18	12,695,475.00 *gal*
Application event totals		0.00	0.00	0.00	50.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.39	11,988,300.00 *gal*
Application event totals		0.00	0.00	0.00	47.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.85	16.92	88.89	526.85	1,340,260.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.26	11,954,625.00 *gal*
Application event totals		61.85	16.92	88.89	574.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.38	16.52	86.78	514.33	1,308,429.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.12	11,920,950.00 *gal*
Application event totals		60.38	16.52	86.78	561.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.21	16.75	87.97	521.40	1,326,404.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.86	11,853,600.00 *gal*
Application event totals		61.21	16.75	87.97	568.26	

5 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.98	16.69	87.64	519.44	1,321,413.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.72	11,819,925.00 *gal*
Application event totals		60.98	16.69	87.64	566.16	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.46	11,752,575.00 *gal*
Application event totals		0.00	0.00	0.00	46.46	

6 - 11/07/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.92	24.21	172.37	923.36	795,360.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.93	5,084,925.00 *gal*
Application event totals		92.92	24.21	172.37	968.29	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.97	24.22	172.47	923.86	795,791.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.20	5,455,350.00 *gal*
Application event totals		92.97	24.22	172.47	972.07	

6 - 05/18/2023: Corn, silage

6 - 05/18/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	60.45	6,841,585.00 *gal*
Application event totals		0.00	0.00	0.00	60.45	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.96	6,785,572.00 *gal*
Application event totals		0.00	0.00	0.00	59.96	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.14	21.11	110.86	657.09	747,820.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.58	6,743,162.00 *gal*
Application event totals		77.14	21.11	110.86	716.68	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.81	19.92	104.64	620.19	705,826.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.96	6,785,572.00 *gal*
Application event totals		72.81	19.92	104.64	680.15	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.26	20.59	108.15	641.03	729,540.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.58	6,743,162.00 *gal*
Application event totals		75.26	20.59	108.15	700.61	

6 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.58	20.41	107.18	635.25	722,955.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.96	6,785,572.00 *gal*
Application event totals		74.58	20.41	107.18	695.20	

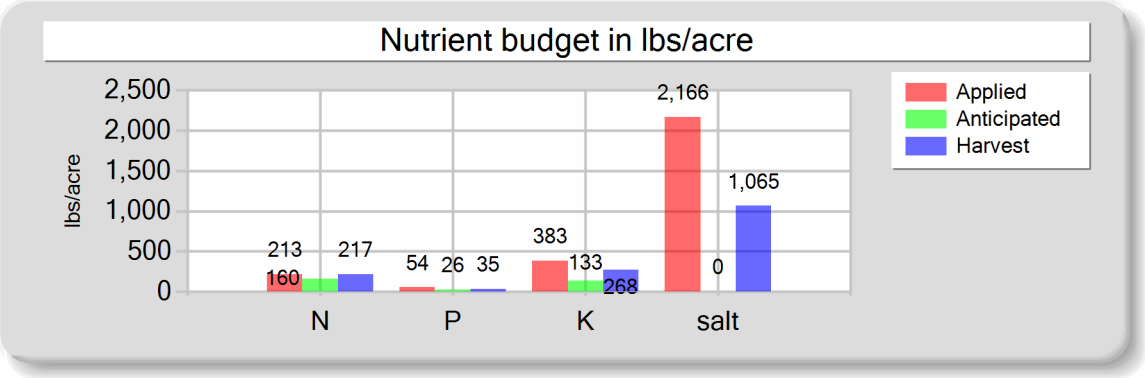
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	58.83	6,658,342.00 *gal*
Application event totals		0.00	0.00	0.00	58.83	

B. NUTRIENT BUDGET

1 - 11/07/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.28	53.74	382.66	2,049.81
Fresh water	0.00	0.00	0.00	116.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.28	53.74	382.66	2,165.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	216.74	35.17	268.07	1,064.68
Nutrient balance	-3.46	18.56	114.59	1,101.20
Applied to removed ratio	0.98	1.53	1.43	2.03

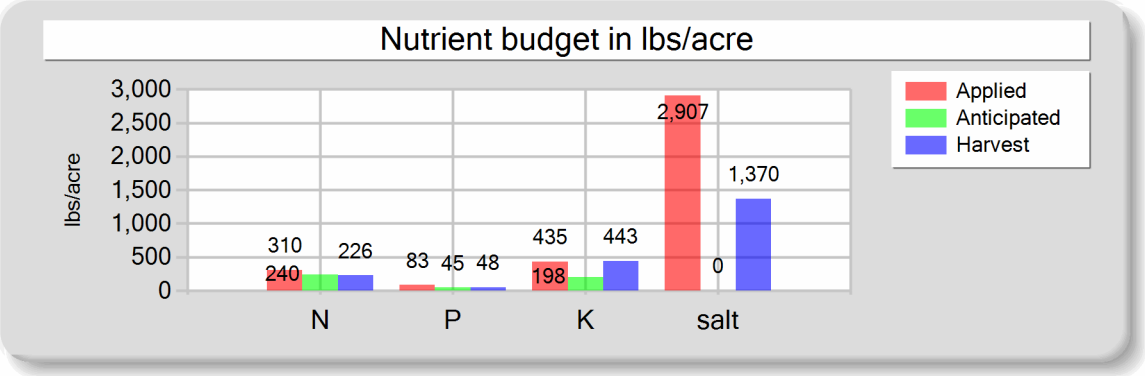
Fresh water applied	
7,341,150.00	*gallons*
270.35	*acre-inches*
14.23	*inches/acre*

Process wastewater applied	
986,688.00	*gallons*
36.34	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/18/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.00	82.90	435.43	2,580.86
Fresh water	0.00	0.00	0.00	326.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.00	82.90	435.43	2,907.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	226.02	48.05	443.14	1,370.34
Nutrient balance	83.98	34.85	-7.71	1,537.13
Applied to removed ratio	1.37	1.73	0.98	2.12

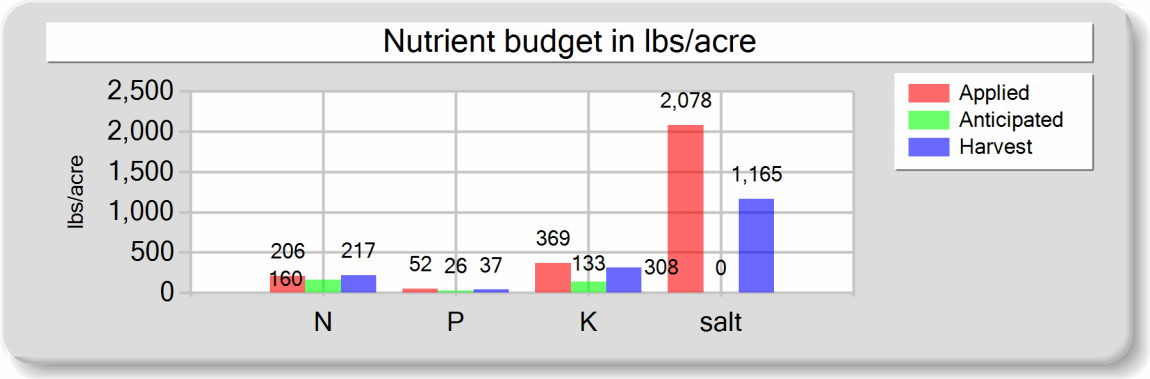
Fresh water applied	
20,656,245.00	*gallons*
760.70	*acre-inches*
40.04	*inches/acre*

Process wastewater applied	
1,641,381.00	*gallons*
60.45	*acre-inches*
3.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/07/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.06	51.86	369.27	1,978.06
Fresh water	0.00	0.00	0.00	100.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.06	51.86	369.27	2,078.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.30	36.74	307.58	1,165.22
Nutrient balance	-11.24	15.11	61.69	913.00
Applied to removed ratio	0.95	1.41	1.20	1.78

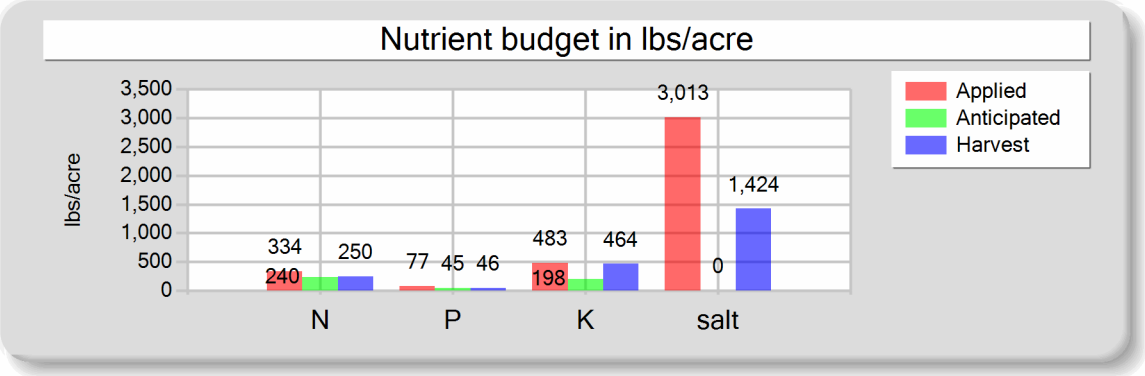
Fresh water applied	
6,667,650.00	*gallons*
245.55	*acre-inches*
12.28	*inches/acre*

Process wastewater applied	
1,002,266.00	*gallons*
36.91	*acre-inches*
1.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/18/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	326.87	77.14	483.46	2,695.32
Fresh water	0.00	0.00	0.00	317.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.87	77.14	483.46	3,012.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	250.17	45.65	463.82	1,424.32
Nutrient balance	83.71	31.49	19.64	1,588.66
Applied to removed ratio	1.33	1.69	1.04	2.12

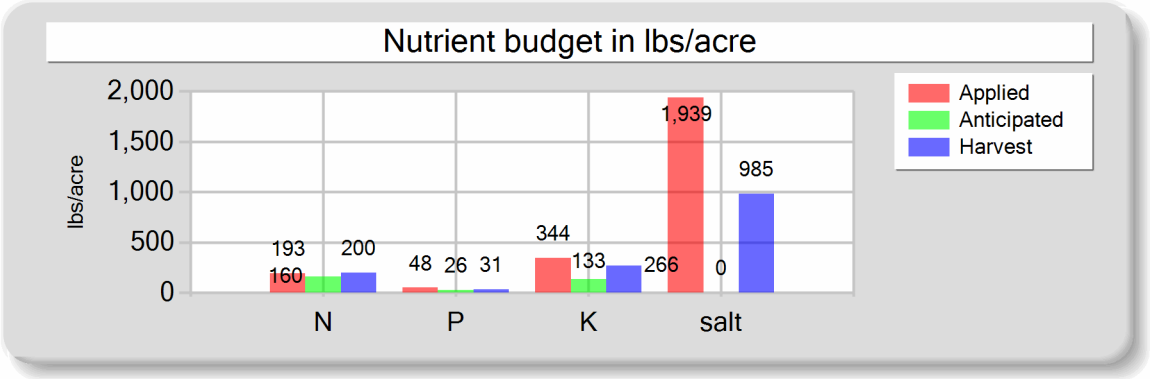
Fresh water applied	
21,147,900.00	*gallons*
778.81	*acre-inches*
38.94	*inches/acre*

Process wastewater applied	
1,818,163.00	*gallons*
66.96	*acre-inches*
3.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/07/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.54	48.33	344.19	1,843.74
Fresh water	0.00	0.00	0.00	95.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.54	48.33	344.19	1,939.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.60	31.38	265.85	984.94
Nutrient balance	-7.06	16.96	78.35	954.40
Applied to removed ratio	0.96	1.54	1.29	1.97

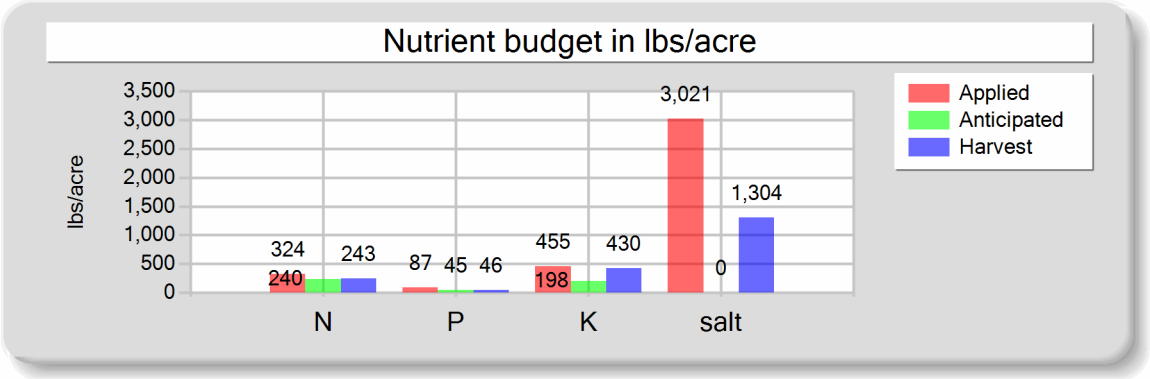
Fresh water applied	
15,591,525.00	*gallons*
574.18	*acre-inches*
11.72	*inches/acre*

Process wastewater applied	
2,288,804.00	*gallons*
84.29	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/18/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	316.76	86.67	455.21	2,698.07
Fresh water	0.00	0.00	0.00	323.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.76	86.67	455.21	3,021.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	243.16	45.81	429.94	1,303.91
Nutrient balance	80.60	40.86	25.27	1,717.27
Applied to removed ratio	1.33	1.89	1.06	2.32

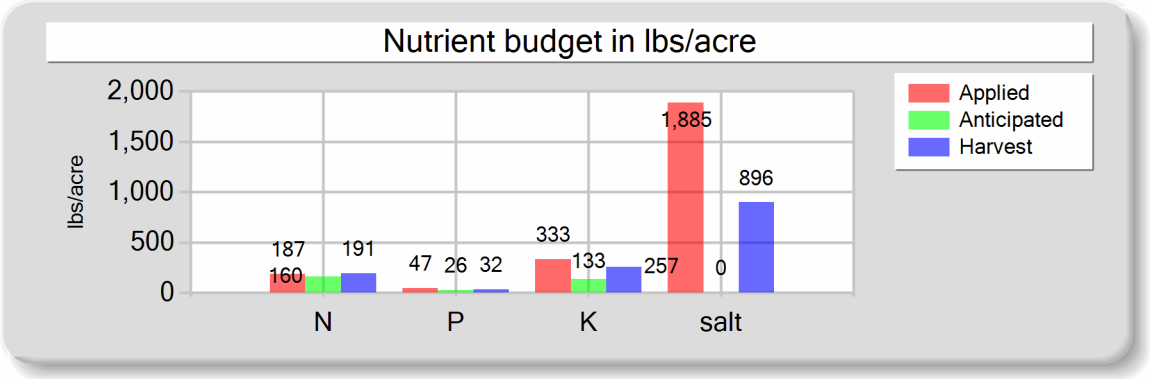
Fresh water applied	
52,701,375.00	*gallons*
1,940.81	*acre-inches*
39.61	*inches/acre*

Process wastewater applied	
4,425,274.00	*gallons*
162.97	*acre-inches*
3.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/07/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	179.61	46.79	333.19	1,784.81
Fresh water	0.00	0.00	0.00	100.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	186.61	46.79	333.19	1,885.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	190.64	31.77	257.44	896.17
Nutrient balance	-4.03	15.02	75.75	989.04
Applied to removed ratio	0.98	1.47	1.29	2.10

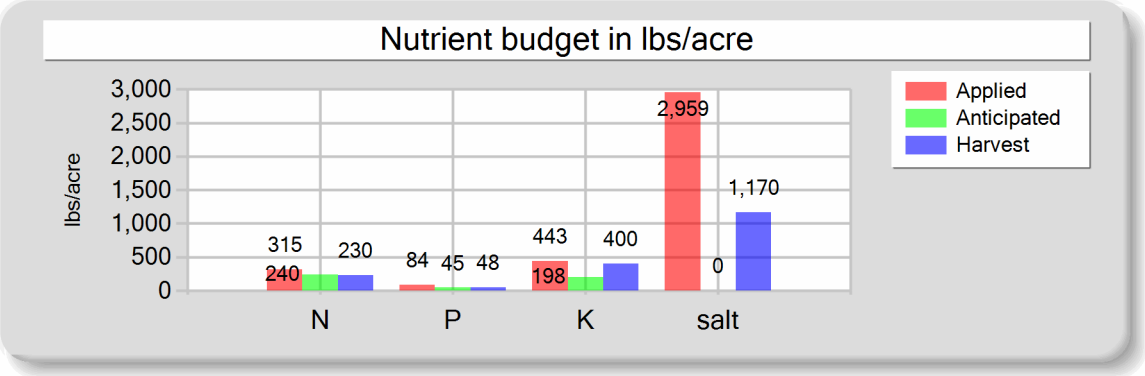
Fresh water applied	
30,745,275.00	*gallons*
1,132.24	*acre-inches*
12.31	*inches/acre*

Process wastewater applied	
4,159,982.00	*gallons*
153.20	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/18/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	308.06	84.29	442.70	2,623.95
Fresh water	0.00	0.00	0.00	335.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.06	84.29	442.70	2,959.34
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.33	47.53	400.33	1,169.91
Nutrient balance	84.73	36.76	42.37	1,789.43
Applied to removed ratio	1.37	1.77	1.11	2.53

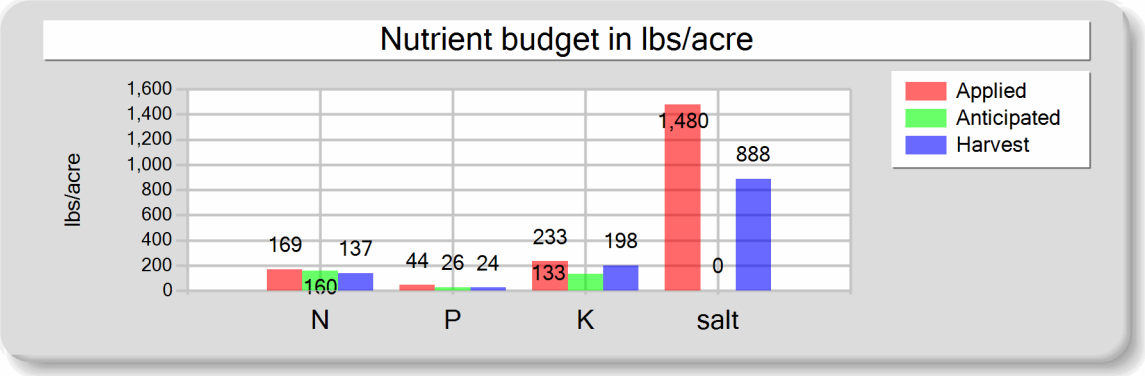
Fresh water applied	
102,708,750.00	*gallons*
3,782.41	*acre-inches*
41.11	*inches/acre*

Process wastewater applied	
8,080,435.00	*gallons*
297.57	*acre-inches*
3.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/07/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.15	44.37	233.02	1,381.13
Fresh water	0.00	0.00	0.00	98.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.15	44.37	233.02	1,479.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.23	23.60	198.08	887.94
Nutrient balance	31.92	20.77	34.94	591.83
Applied to removed ratio	1.23	1.88	1.18	1.67

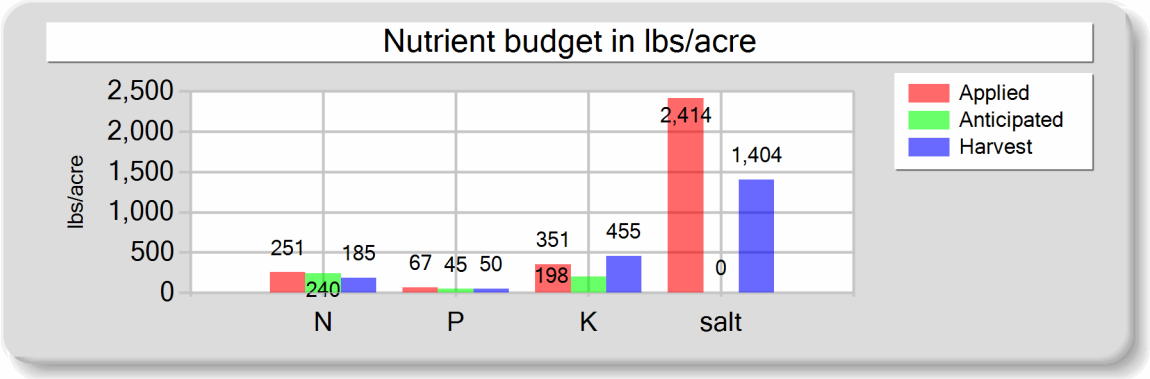
Fresh water applied	
24,953,175.00	*gallons*
918.94	*acre-inches*
12.09	*inches/acre*

Process wastewater applied	
3,513,492.00	*gallons*
129.39	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/18/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	244.43	66.88	351.27	2,082.02
Fresh water	0.00	0.00	0.00	331.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.43	66.88	351.27	2,414.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	185.42	50.40	455.44	1,404.11
Nutrient balance	66.02	16.48	-104.17	1,009.89
Applied to removed ratio	1.36	1.33	0.77	1.72

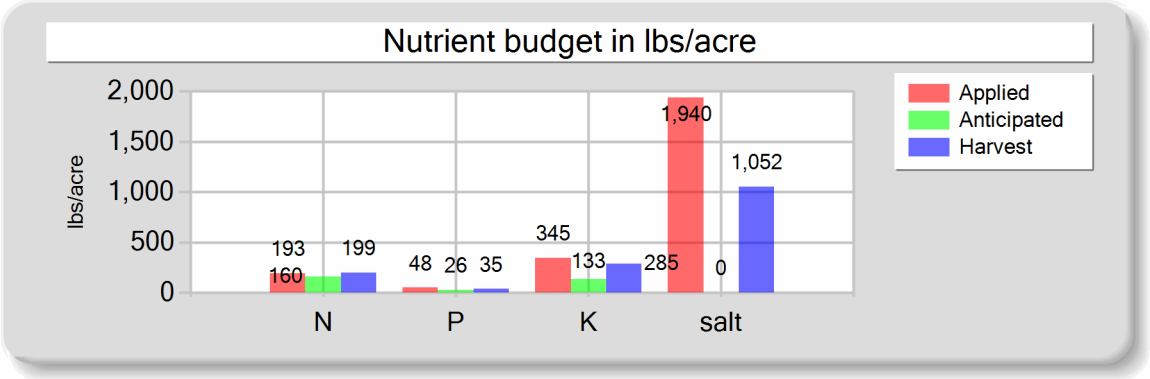
Fresh water applied	
83,985,450.00	*gallons*
3,092.90	*acre-inches*
40.70	*inches/acre*

Process wastewater applied	
5,296,506.00	*gallons*
195.05	*acre-inches*
2.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/07/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.89	48.43	344.84	1,847.23
Fresh water	0.00	0.00	0.00	93.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.89	48.43	344.84	1,940.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.24	35.40	285.21	1,051.84
Nutrient balance	-6.35	13.03	59.63	888.51
Applied to removed ratio	0.97	1.37	1.21	1.84

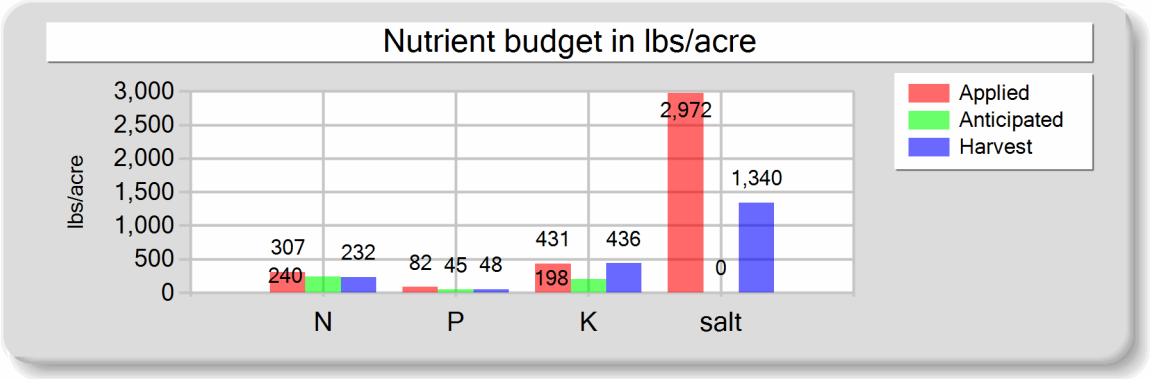
Fresh water applied	
10,540,275.00	*gallons*
388.16	*acre-inches*
11.42	*inches/acre*

Process wastewater applied	
1,591,151.00	*gallons*
58.60	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 05/18/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	299.79	82.03	430.83	2,553.57
Fresh water	0.00	0.00	0.00	418.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.79	82.03	430.83	2,971.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	232.34	48.26	436.09	1,340.43
Nutrient balance	74.45	33.77	-5.26	1,631.45
Applied to removed ratio	1.32	1.70	0.99	2.22

Fresh water applied	
47,342,967.00	*gallons*
1,743.48	*acre-inches*
51.28	*inches/acre*

Process wastewater applied	
2,906,141.00	*gallons*
107.02	*acre-inches*
3.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 06/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 3.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	4,500.00	22,300.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,400.00	7,600.00	24,800.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,180.00	465.00	0.00	1.90	204.00	912.00								9,590.00	4,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Waste water

Sample and source description: Waste water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	462.00	368.00	0.00	1.20	61.00	739.00								7,930.00	3,470
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	420.00	304.00	0.00	0.30	115.00	604.00								7,370.00	3,580
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	476.00	353.00	0.00	0.00	124.00	883.00								8,870.00	4,730
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well #1

Domestic Well - Well #1

Well #1

Sample description: Well #1

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.00	0.00	4.00								1,780.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - Well #2

Well #2

Sample description: Well #2

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.10	0.00	4.10								1,800.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - Well #3

Well #3

Sample description: Well #3

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90	0.00	3.90								1,780.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - Well #4

Domestic Well - Well #4

Well #4

Sample description: Well #4

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	21.50	0.00	21.50								2,740.00	
DL	0.10	0.10	0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								32.50	36
DL	0.10	0.10	0.10								10.00	10

D. SOIL ANALYSES

1

Field 1

Sample and source description: Field 1

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value	5.29		54.00				
DL	1.00		1.00				

2

Field 2

Sample and source description: Field 2

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value	19.50		106.00	398.00			
DL	1.00		1.00	1.00			

3

Field 3

Sample and source description: Field 3

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value	20.10		96.00	304.00			
DL	1.00		1.00	1.00			

4

Field 4

Sample and source description: Field 4

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value	9.36		42.00	298.00			
DL	1.00		1.00	1.00			

5

5

Field 5

Sample and source description: Field 5

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value	19.90		94.00	291.00			
DL	1.00		1.00	1.00			

E. PLANT TISSUE ANALYSES

1 - 11/07/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,800.00	3,700.00	28,200.00		11.20
DL	100.00	100.00	100.00		0.01

1 - 05/18/2023: Corn, silage

1

Sample and source description: 1

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,700.00	24,900.00		7.70
DL	100.00	100.00	100.00		0.01

2 - 11/07/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	3,500.00	29,300.00		11.10
DL	100.00	100.00	100.00		0.01

2 - 05/18/2023: Corn, silage

2

Sample and source description: 2

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,500.00	25,400.00		7.80
DL	100.00	100.00	100.00		0.01

3 - 11/07/2022: Oats, silage-soft dough

3A

Sample and source description: 3A

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	3,600.00	30,500.00		11.30
DL	100.00	100.00	100.00		0.01

3 - 11/07/2022: Oats, silage-soft dough

3B

Sample and source description: 3B

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	3,700.00	29,400.00		15.70
DL	100.00	100.00	100.00		0.01

3 - 05/18/2023: Corn, silage

3A

Sample and source description: 3A

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,600.00	24,400.00		7.40
DL	100.00	100.00	100.00		0.01

3B

Sample and source description: 3B

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,600.00	25,900.00		7.80
DL	100.00	100.00	100.00		0.01

4 - 11/07/2022: Oats, silage-soft dough

4 - 11/07/2022: Oats, silage-soft dough

4A

Sample and source description: 4A

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	3,700.00	29,200.00		12.70
DL	100.00	100.00	100.00		0.01

4B

Sample and source description: 4B

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,300.00	3,800.00	31,100.00		10.40
DL	100.00	100.00	100.00		0.01

4C

Sample and source description: 4C

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,400.00	3,900.00	31,600.00		11.00
DL	100.00	100.00	100.00		0.01

4 - 05/18/2023: Corn, silage

4 - 05/18/2023: Corn, silage

4A

Sample and source description: 4A

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,000.00	23,100.00		6.60
DL	100.00	100.00	100.00		0.01

4B

Sample and source description: 4B

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,600.00	21,900.00		6.40
DL	100.00	100.00	100.00		0.01

4C

Sample and source description: 4C

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,600.00	21,900.00		7.20
DL	100.00	100.00	100.00		0.01

5 - 11/07/2022: Oats, silage-soft dough

5 - 11/07/2022: Oats, silage-soft dough

5A

Sample and source description: 5A

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	3,800.00	31,900.00		14.30
DL	100.00	100.00	100.00		0.01

5B

Sample and source description: 5B

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	3,900.00	32,600.00		13.50
DL	100.00	100.00	100.00		0.01

5 - 05/18/2023: Corn, silage

5A

Sample and source description: 5A

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,700.00	23,700.00		7.40
DL	100.00	100.00	100.00		0.01

5 - 05/18/2023: Corn, silage

5B

Sample and source description: 5B

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	2,800.00	25,300.00		7.80
DL	100.00	100.00	100.00		0.01

6 - 11/07/2022: Oats, silage-soft dough

6

Sample and source description: 6

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	3,500.00	28,200.00		10.40
DL	100.00	100.00	100.00		0.01

6 - 05/18/2023: Corn, silage

6

Sample and source description: 6

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,700.00	24,400.00		7.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

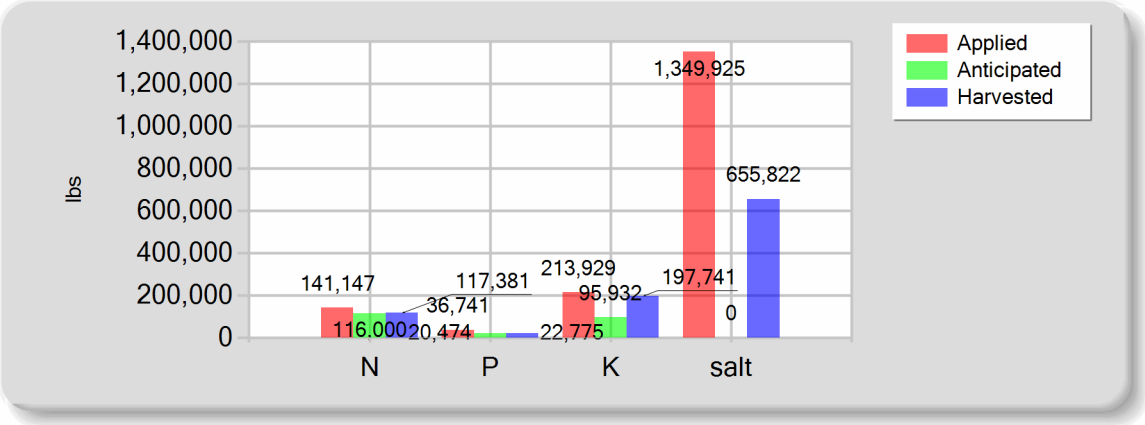
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

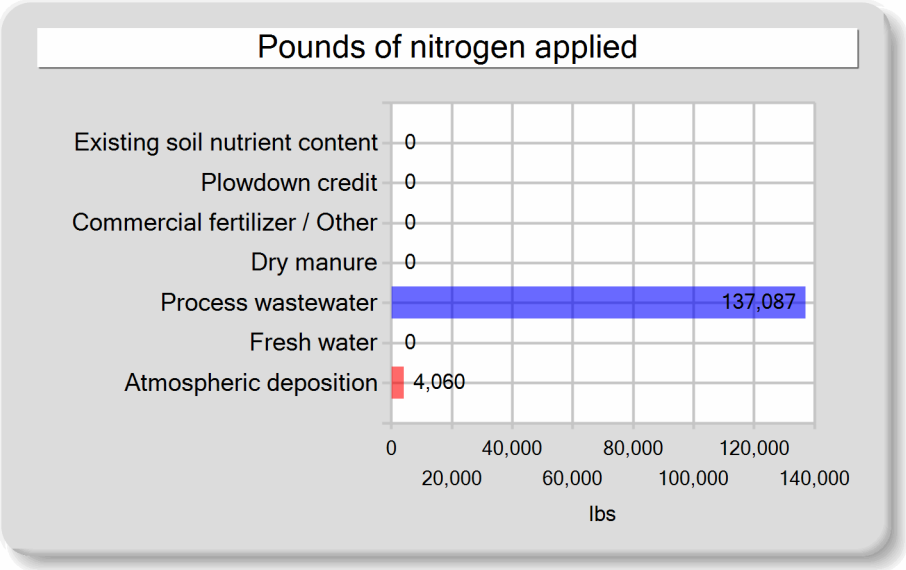
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

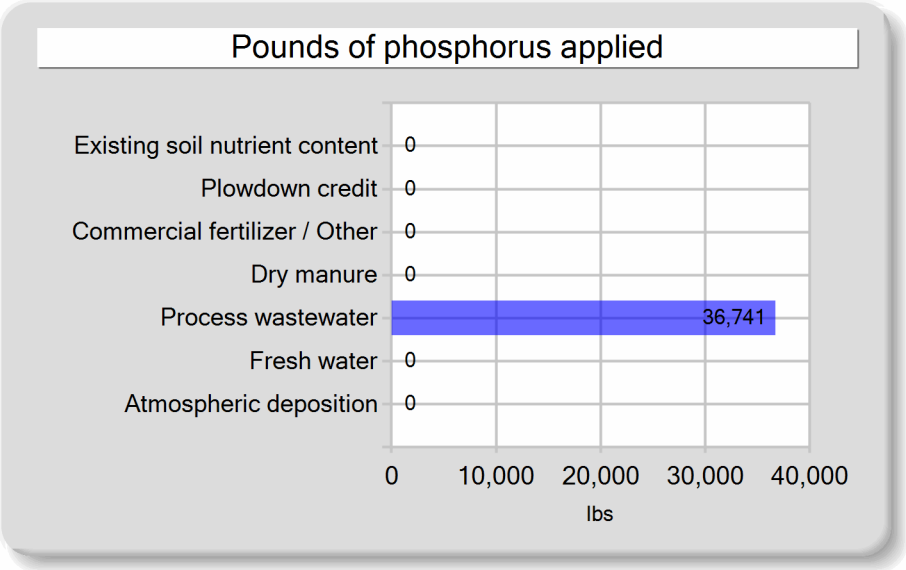
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	137,087.24	36,740.85	213,928.94	1,222,431.78
Fresh water	0.00	0.00	0.00	127,492.76
Atmospheric deposition	4,060.00	0.00	0.00	0.00
Total nutrients applied	141,147.24	36,740.85	213,928.94	1,349,924.54
Anticipated crop nutrient removal	116,000.00	20,474.00	95,932.00	0.00
Actual crop nutrient removal	117,380.59	22,775.37	197,740.65	655,822.18
Nutrient balance	23,766.65	13,965.48	16,188.29	694,102.37
Applied to removed ratio	1.20	1.61	1.08	2.06

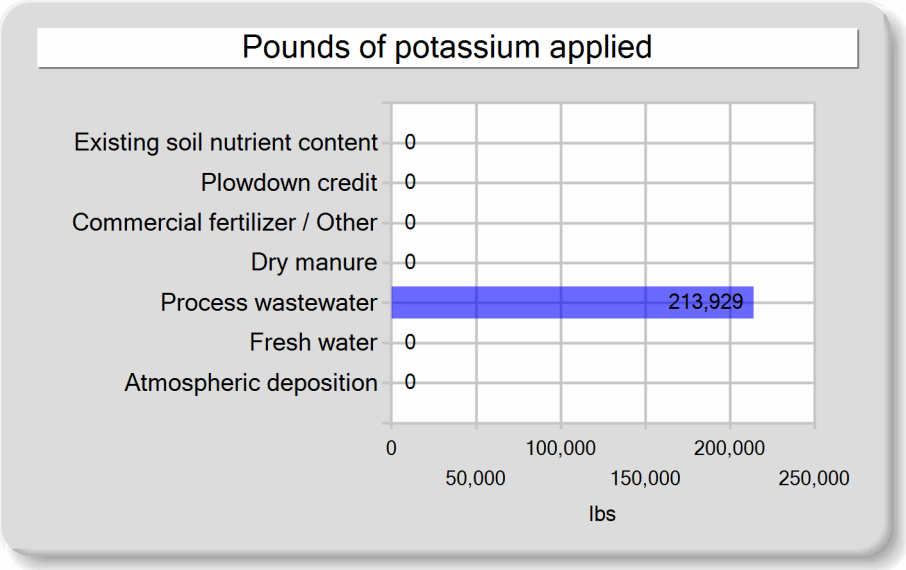
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

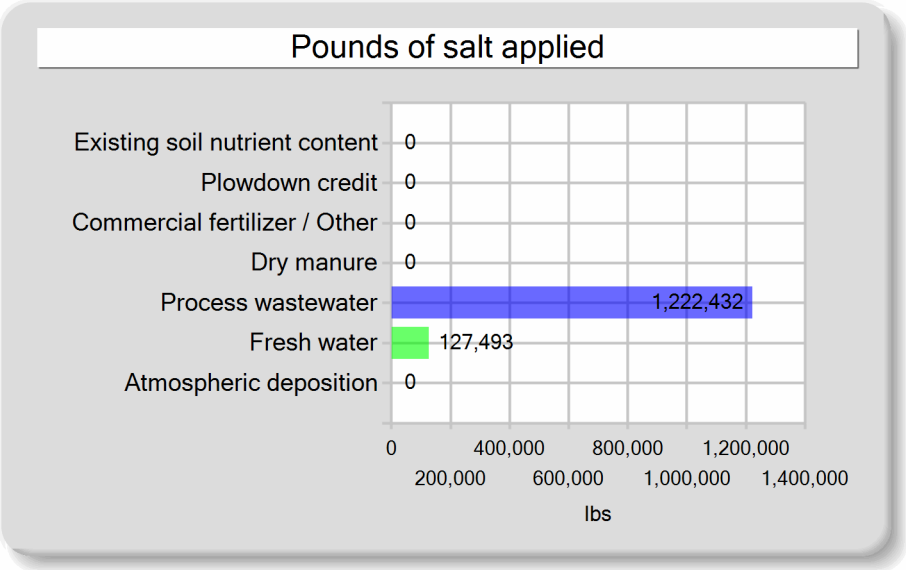


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Arie Roest
Arie Roest (Jun 19, 2024 16:39 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Arie Roest	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/19/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0895-01	Well #1	Ag Water	T. Madruga		11/15/2023 13:00
23K0895-02	Well #2	Ag Water	T. Madruga		11/15/2023 13:30
23K0895-03	Well #3	Ag Water	T. Madruga		11/15/2023 14:00
23K0895-04	Well #4	Ag Water	T. Madruga		11/15/2023 14:30

Default Cooler Temperature on Receipt °C: 0.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Sample Results

Sample: Well #1
23K0895-01 (Water)

Sampled: 11/15/2023 13:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.78**	mmhos/cm	0.01	1		11/20/23 13:34	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1780**	umhos/cm	10.0	1		11/20/23 13:34	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 13:00	Field		BEK0554
Nitrate Nitrogen as NO3N	**4.0**	mg/L	0.1	1	10	11/16/23 19:11	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:34	SM 4500-H+	H	BEK0616
pH	**7.5**	units	1.0	1		11/20/23 13:34	SM 4500-H+	H	BEK0616

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Sample Results
(Continued)

Sample: Well #2
23K0895-02 (Water)

Sampled: 11/15/2023 13:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.80**	mmhos/cm	0.01	1		11/20/23 13:36	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1800**	umhos/cm	10.0	1		11/20/23 13:36	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 13:30	Field		BEK0554
Nitrate Nitrogen as NO3N	**4.1**	mg/L	0.1	1	10	11/16/23 19:31	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:36	SM 4500-H+	H	BEK0616
pH	**7.6**	units	1.0	1		11/20/23 13:36	SM 4500-H+	H	BEK0616

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Sample Results
(Continued)

Sample: Well #3
23K0895-03 (Water)

Sampled: 11/15/2023 14:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.78**	mmhos/cm	0.01	1		11/20/23 13:37	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1780**	umhos/cm	10.0	1		11/20/23 13:37	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 14:00	Field		BEK0554
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	11/16/23 19:51	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:37	SM 4500-H+	H	BEK0616
pH	**7.6**	units	1.0	1		11/20/23 13:37	SM 4500-H+	H	BEK0616

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Sample Results
(Continued)

Sample: Well #4
23K0895-04 (Water)

Sampled: 11/15/2023 14:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.74**	mmhos/cm	0.01	1		11/20/23 13:39	SM 2510 B		BEK0616
Electrical Conductivity umhos	**2740**	umhos/cm	10.0	1		11/20/23 13:39	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 14:30	Field		BEK0554
Nitrate Nitrogen as NO3N	**21.5**	mg/L	0.1	1	10	11/16/23 20:11	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:39	SM 4500-H+	H	BEK0616
pH	**7.4**	units	1.0	1		11/20/23 13:39	SM 4500-H+	H	BEK0616

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DELLAVALLE™ LABORATORY INC

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8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573										
Blank (BEK0573-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK4)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0573-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0573-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEK0573-BS3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEK0573-DUP1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.173	10
Duplicate (BEK0573-DUP2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.629	10
Duplicate (BEK0573-DUP3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	1.5		0.1	mg/L		1.5			0.195	10
Matrix Spike (BEK0573-MS1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	11.0		0.1	mg/L	5.000	5.8	104	90-110		
Matrix Spike (BEK0573-MS2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.3	97.1	90-110		
Matrix Spike (BEK0573-MS3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.5	106	90-110		
Reference (BEK0573-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0573-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573 (Continued)										
Reference (BEK0573-SRM3)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0573-SRM4)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616										
Blank (BEK0616-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0616-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.2		1.0	units						
Blank (BEK0616-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.7		1.0	units						
Duplicate (BEK0616-DUP1)		**Source: 23K0896-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.47		0.01	mmhos/cm		0.47			0.423	10
pH	7.8		1.0	units		7.8			0.255	10
Electrical Conductivity umhos	472		10.0	umhos/cm		474			0.423	10
Duplicate (BEK0616-DUP2)		**Source: 23K0899-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.65		0.01	mmhos/cm		1.64			0.298	10
Electrical Conductivity umhos	1650		10.0	umhos/cm		1640			0.298	10
pH	7.0		1.0	units		7.0			0.00	10
Reference (BEK0616-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	561			umhos/cm	538.0		104	90-110		
Reference (BEK0616-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0616-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0616-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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LABORATORY INC

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Hilmar, CA 95324

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Roestar Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:04

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616 (Continued)										
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0616-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/16/23 15:35

23K0895

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 4 No. of Bottles: 4

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Roestar Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1	11/15/2023	1300	.1	ND
2	Well #2	11/15/2023	1330	.7	ND
3	Well #3	11/15/2023	1400	.8	ND
4	Well #4	11/15/2023	1430	2.1	ND
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1430	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11/16/23 3:25
Third		DLI	11/16/23 15:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0895

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice:☑ Blue Ice:☐ None:☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1						
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										